

SANJOY KUMAR

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Education

Texas Tech University

Doctor of Philosophy (Ph.D.) in Applied Mathematics (Engineering Math)

Research Areas: Machine Learning, Computer Vision, Physical AI, Robotics

2026-Present

Completing 14 Credits

Shahjalal University of Science & Technology

Bachelor of Science in Electrical & Electronic Engineering

Research Areas: Machine Learning, Computer Vision, Signal & Image Processing, Robotics & Autonomous System

2016-2020

CGPA: 3.56/4.00

Research Experience

Research Assistant • High-Performance Computing Center, Texas Tech University

Apr'2024-Present

Supervisor: Dr Yifan Wang, Assistant Professor, Dept of Mathematics & Statistics, Texas Tech University

- Build a complete pipeline that treats a **hyperuniform micropost microfluidic device as a biomechanical encoder** immersed-boundary simulations generate size- and elasticity-dependent cell trajectories, classified by sequence models.
- Developed hybrid physics-informed sequence models that embed fluid-structure and PDE residuals directly into the network, enabling accurate multi-step trajectory & simultaneous prediction of morphological descriptors (elasticity, shape deformation, aspect ratio).
- Established a clear design principle for **passive microfluidic separators: cell size drives the dominant**, history-dependent lateral separation, elasticity acts as a secondary modulator providing concrete guidance **for optimizing label-free rare-cell isolation devices**.
- Created a transferable SciML framework that couples multiphysics simulation with data-driven surrogates, real-time trajectory-based phenotyping for **diagnostic microfluidics, biomedical sensing, microrobotic systems interacting with deformable cells**.

[\[Paper-1\]](#) [\[Paper-2\]](#) [\[Paper-3\]](#) [\[Paper-4\]](#)

Research Collaborator • Center for Research & Innovation, RPSU

Sep' 2025-' Aug'2026

Supervisor: Shovan Kumar Paul, Assistant Professor, Dept of Computer Science & Engineering, R.P. Shaha University, BD

- Developed a **Physics-Informed Neural Dynamics (PIND) architecture that combines analytical multi-body rigid-body models** with structured residual networks for actuator compliance, soft contact forces, trained under momentum conservation, friction-cone.
- Embedded the continuous-time learned model into a real-time **Nonlinear Model Predictive Control (NMPC) framework** solved Seq Quadratic Programming, achieving closed-loop locomotion tracking on the Unitree Go1 using only 10% of the training data.
- Designed **Port-Hamiltonian residual network that structurally guarantees mechanical passivity and coupled** with High-Order Control Barrier Func, adaptive conformal prediction to create quadratic-program safety filter for reinforcement learning policies.
- Established a unified approach for data-efficient dynamics learning and certified safe control that bridges physics-informed machine learning with model-based predictive control, **safety-critical reinforcement learning for legged and contact-rich robotic systems**.

[\[Paper-1\]](#) [\[Paper-2\]](#)

Research Assistant • Brandon University

Dec' 2023-' Oct'2025

Supervisor: Dr Gautam Srivastava, Professor, Dept of Computer Science & Mathematics, Brandon University, Canada

- Developed a unified multimodal fusion architecture that **jointly processes visual, textual, and behavioral streams** via a learned cross-modal attention mechanism, analogous to sensor-fusion pipelines used in mobile robots and autonomous systems.
- Introduced a **GRU-based neural narrative tracker with specialized memory banks** that maintains long-horizon conversational state, enabling coherent multi-turn decision making under evolving user intent.
- Established a reusable architecture pattern that combines LLM reasoning, formal optimization objectives, and human-centered interfaces, enabling scalable deployment of adaptive AI systems in data-intensive and safety-sensitive environments.

[\[Paper-1\]](#) [\[Paper-2\]](#) [\[Paper-3\]](#)

Research Collaborator • Florida International University

Sep' 2021-Feb' 2022

Supervisor: Dr Naphtali Rishe, Professor, Dept of Computer Science, Florida International University

- Investigated **deep learning-based image reconstruction and computer vision methods** for detecting and removing cloud-affected regions in high-resolution satellite imagery.
- Explored **Deep Image Prior and convolutional neural network architectures** for unsupervised image restoration, addressing image denoising, inpainting, and reconstruction without requiring large, labeled training datasets.
- Analyzed **spatial, spectral, and multisource remote-sensing information** for robust visual perception and image reconstruction, with applications to intelligent sensing and autonomous vision systems.

[\[Paper-1\]](#)

Research Assistant • Shahjalal University of Science & Technology

May' 2020-Feb' 2021

Supervisor: Shahid Iqbal, Assistant Professor, Dept of Electrical Engineering, Shahjalal University of Science & Technology

- Developed as **IoT-based remote health monitoring system** integrating NTC temperature and MAX30102 heart-rate/SpO2 Sensors with EXP32 microcontroller for continuous patient monitoring.
- Designed real-time data accusation and transmission pipeline using WIFI/ESP32 to process, transmit and store on server.
- Implemented web-based visual and patient management interface enabling physicians to remotely monitor records.

[\[Paper-1\]](#)

Professional Experience

Sr Machine Learning Engineer • Bista Solutions Inc, USA (Bangladesh Office)

Apr' 2024 – Aug' 2026

- **Architected and optimized production-scale AI systems (LLM and visionLLM)** using applying parameter-efficient fine-tuning, quantization, inference optimization to develop domain-adapted intelligent systems for enterprise decision-support and automation.
- **Developed computer-vision and multimodal AI solutions** integrating image, text, structured data for automated visual analysis, recognition, classification, inspection workflows; deployed vision-driven automation for enterprise and manufacturing processes.
- Designed **multi-agent intelligent systems** combining multimodal perception, retrieval-augmented generation, high-dimensional embeddings, tool execution, and workflow orchestration to transform complex multi-step processes into automated AI-driven decision.
- **Engineered ML/GenAI-based manufacturing automation systems**, integrating computer vision, ML, and enterprise data to automate industrial workflows, process analysis, operational decision support; led end-to-end development & production deployment.

Sr Machine Learning Engineer • DataSense, iSocial Limited

Feb' 2022 – Mar' 2024

- **Researched and fine-tuned multilingual transformer architectures** using domain-specific datasets, achieving **88% classification accuracy** across seven languages and integrating the resulting models into production conversational AI systems.
- **Developed computer vision and deep learning solutions** for automated image analysis, incorporating convolutional neural networks and representation-learning techniques for visual classification and intelligent data-processing workflows.
- **Engineered and deployed production MLOps infrastructure** using AWS SageMaker, MLflow, and DVC for reproducible experimentation, model registry, data versioning, and scalable model deployment.

Machine Learning Engineer • IntelSense AI Ltd, Bangladesh

Apr' 2021 – Jan' 2022

- **Engineered end-to-end machine learning and computer vision pipelines** spanning data acquisition, image preprocessing, feature extraction, model development, optimization, and performance analysis; identified model failure.
- **Designed and deployed edge-based computer vision workflows on NVIDIA Jetson**, integrating deep learning inference with image acquisition and preprocessing to enable computationally efficient, hardware-aware intelligent vision applications.
- **Architected a hardware–software data acquisition and automation pipeline** integrating handheld scanners, imaging devices, and printers with ML/CV processing components; automated device-data retrieval, visual/document analysis.

Data Scientist • PIPELINE Security Ltd, Japan (Bangladesh Office)

May' 2020 – Mar' 2021

- **Optimized end-to-end computational and ML workflows**, identifying system bottlenecks and redundant processing stages to achieve a **12% improvement in overall project efficiency**.
- **Designed and implemented a real-time hardware data-acquisition pipeline** for collecting and processing measurements from **multiple heterogeneous sensors**, establishing the foundation for sensor-driven intelligent systems and automated decision-making.
- Developed and evaluated **machine learning pipelines spanning regression, classification** complex sensor and operational datasets.

Teaching Experience

Graduate Assistantship (TA) • Texas Tech University

Sep'2026 – Present

- Support undergraduate mathematics instruction through tutoring, problem-solving, and student mentoring.
- Assist with course-related academic activities and provide guidance in mathematical concepts and computational methods.
- Facilitate students' understanding of mathematical and computational concepts through one-on-one and small-group instruction, fostering analytical and problem-solving skills.

Instructor (Machine Learning Course) • Infinity Flame Soft Institute, Bangladesh

Sep' 2024 – June' 2021

- Taught **150+ students** in machine learning, covering supervised and unsupervised learning, model evaluation, feature engineering, and practical ML workflows using **Python and Scikit-learn**.
- Delivered hands-on instruction through coding exercises, assignments, and real-world projects, guiding students in building, training, evaluating, and improving machine learning models.

Mentor • SEC Robotics Lab, SEC, SUST

Apr' 2019 – Aug' 2020

- Mentored students in **robotics, embedded systems, and programming**, providing hands-on guidance in designing, building, and testing robotic systems.
- Guided student teams in **microcontroller programming, sensor integration, motor control, and hardware–software interfacing** for project developments.

Peer Reviewing Experience

[1] Journal of Cancer and Tumor International [[Certificate1](#)] [[Certificate2](#)] [[Certificate3](#)] [[Certificate4](#)]

[2] Archives of Current Research [[Certificate](#)]

[3] Asian Journal of Case Report in Surgery [[Certificate](#)]

[4] Asian Journal of Research in Surgery [[Certificate](#)]

[5] Asian Oncology Research Journal [[Certificate](#)]

[6] Book Chapter Review: *Clinicopathological Profile of Breast Cancer in India: A Five-year Retrospective Case Series*. [[Certificate](#)]

[7] Book Chapter Review: *Breast Carcinoma: A Global Perspective on Prevalence and Impact*. [[Certificate](#)]

Publications

- **Sanjoy Kumar** and Yifan Wang, “Physics-Informed Fourier Basis Neural Networks for PDEs: Architecture, Analysis, and Applications to Heat, Allen-Cahn, Burgers’ and Kovasznay Flow Equations”, *International Journal of Computer Mathematics*, 2026. [2nd Reviewer Feedback Submitted] [[Link](#)]
- **Sanjoy Kumar**, Shovan Kumar Paul, Md Monir Hossain, “Port-Hamiltonian Residual Learning with Adaptive Conformal Barrier Filters for Safe Robot Reinforcement Learning”, *2026 8th International Conference on Control and Robotics, Japan*. [Peer Reviewed] [[Link](#)]
- **Sanjoy Kumar**, Shovan Kumar Paul, Lula Babole Alphonse-Roger “Physics-Informed Residual Neural Dynamics for Data-Efficient Predictive Control in Legged Robots”, *2026 2nd International Conference on Advanced Computing & Intelligent Robotics Applications*. [Peer Reviewed] [[Link](#)]
- **Sanjoy Kumar**, Huixin Zhan, Karl Gardner, Travis Thompson, Wei Kei, Yifan Wang “Classification of Circulating Tumor Cells Using Machine Learning on Microfluidic Trajectory Data”, *2025 17th International Conference on Machine Learning and Computing, Paper ID 4170, 2024, J. ACM, Vol. 37, No. 4, Article 111*. [[Link](#)]
- Rifat Rejuan; Eugenio Aulisa; Wei Li; Travis Thompson; **Sanjoy Kumar**; Suncica Canic; Yifan Wang, “Validation of a Microfluidic Device Prototype for Cancer Detection and Identification: Circulating Tumor Cells Classification Based on Cell Trajectory Analysis Leveraging Cell-Based Modeling and Machine Learning”, *International Journal for Numerical Methods in Biomedical Engineering, Volume 41, Issue 4, 2025*. [[Link](#)]
- Egharevba, Lawrence; **Kumar, Sanjoy**; Amini, Hadi; Adjouadi, Malek; Rishe, Naphtali "Detecting and Removing Clouds Affected Regions from Satellite Images Using Deep Learning", *IPSI Transactions on Internet Research*, vol. 19(2), pp. 13-23, 2023. [[Link](#)]
- Md. Irfanul Hasan Tusar¹*, **Sanjoy Kumar**², K.M.E. Hasan³ and Md. Shahid Iqbal⁴ "IoT Based Health Monitoring System in Case of Older & Infectious Diseases in Bangladesh." *6th International Conference on Engineering Research, Innovation and Education School of Applied sciences & Technology, SUST, Sylhet, ICEIRE 2021* [[Link](#)]
- **Sanjoy Kumar** and Gautam Srivastava, “AXAI-MH: An Adaptive Context-Sensitive Explainable AI Framework for Personalized Mental Health Risk Assessment.” *IEEE Transactions on Computational Social Systems, Paper No. TCSS-2025-07-1320*. [[Link](#)]
- **Sanjoy Kumar** and Gautam Srivastava, “Multimodal Knowledge Augmented Narrative Driven Unified Recommendation System.” *IEEE Transactions on Computational Social Systems, Paper No. TCSS-2025-06-1183*. [Peer Reviewed] [[Link](#)]
- **Sanjoy Kumar** and Gautam Srivastava, Senior Member, IEEE, “AutoMaterialize: LLM-Powered Automated Materialized View with Schema-Guided in-Context Learning Optimization for Analytical Workloads. [[Link](#)]
- **Sanjoy Kumar**, “DeepSQL-R1: A Quantized LLM for High-Performance and Reinforcement Driven NL2SQL Generation”, *figshare. Journal contribution. https://doi.org/10.6084/m9.figshare.28330301.v1* [[Link](#)]

Research Report

- Dr Ananya Raihan, Asif Ayon, **Sanjoy Kumar** "Research Report: Skills and Employments for the Future in Bangladesh." *Economic Intelligence Bangladesh, Powered by The Business Standard*, 2024 [[Link](#)]
- Report: State of Work in Bangladesh Gig Economy & Digital labor Platforms. Fairwork Bangladesh Ratings 2022 Report, 2023 [[Link](#)]
Collaboration: University of Oxford (Oxford Internet Institute), DataSense (iSocial Limited, Bangladesh)
- **Sanjoy Kumar**, “Research Project: Yield Curve as Recession Indicator in the Framework of Machine Learning.” #2,173 in *Education Theory Research, ASIN: B09BLBYBBG, Amazon Kindle Publisher*, 2022 [[Link](#)]
- Sanjoy Kumar, Performance Improvement of a Diesel Engine Power Plant with Optimal Utilization. ASIN: B082VJCXDL, Amazon Kindle Publisher, 2023 [[Link](#)]

Books Publication

- [1] Arduino-Based Embedded Systems: Development and Maintenance, 2018 [[Link](#)], **Author:** Sanjoy Kumar, **ASIN:** B081G9RJPL
- [2] Device Integrated Dart Programming Language for Beginner, 2019 [[Link](#)], **Author:** Sanjoy Kumar, **ASIN:** B08174FM7Q
- [3] GIT: A Distributed Version Control System, 2019 [[Link](#)], **Author:** Sanjoy Kumar, **ASIN:** B083TZR9KK

Skills

- **Programming:** Python, C++, Matlab, Verilog, VHDL
- **AI & Machine Learning:** PyTorch, Tensorflow, JAX, Keras, Scikit-learn, Deep learning, Reinforcement learning, MLOps
- **Hardware & Embedded System:** Microcontrollers, PCB Design (SPICE/Altium), Peripheral Interfacing, NVIDIA Jetson, Hardware-Software Integration, CUDA, GPU Computing, Sensor Interface Design, Embedded AI
- **Robotics Technique & Simulation:** ROS 2, Gazebo, NVIDIA ISAAC Sim, PCL, 3D Computer Vision, Control theory, Sensor Fusion, State Estimation, Motion Planning, Model Predictive Control, kalman Filtering
- **Computer Vision & Signal Processing:** OpenCV, Open3D, Image Processing, Signal Processing, Point-Cloud Processing
- **Scientific Computing:** Physics-Informed Neural Network, DeepONET, Neural ODE, Differentiable Physics, PDEs, FEniCS
- **Generative AI:** LLM fine-tuning, RAG, Multi-agent workflow, Langchain, MCP, LLMops, Scale AI Product, Agent Deployment
- **Tools:** LaTeX, Linux OS, ParaView, Git/Github, Docker
- **Functional Skills:** Project Management, Research & Facilitations, Documentation, Report Writing, Research Proposal Writing

Entrepreneurship & Startup Experience

Movmi: AI Powered 3D Motion Capture and Animation Creator [\[Link\]](#), Accepted by NVIDIA Inception & Microsoft Startups Program
Role: Co-founder & Chief Technology Officer (CTO), 2020-2024, **Total Users:** 12000+,

Scholyx: AI Lens for Research Automation [Acquired by Ilapata Tech Inc] [\[Link\]](#)
Role: Co-founder & Chief Executive Officer (CEO), 2024-2026, **Total Users:** 18000+

Leadership & Extracurricular

Program Operator , <i>International Labor Organization, Bangladesh Gig Economy, DataSense, iSocial Limited</i>	2023
Program Research Executive , <i>SEC Robotics Lab, Sylhet Engineering College, SUST</i>	2020-2021
Program Coordinator (Sponsors, Events) , <i>SEC Robotics & Tech Contest Program, Dept of EEE, SEC, SUST</i>	2018-2019
Program Coordinator , <i>Inter University Programming Contest, SEC, SUST</i>	2017
Scout Leader (School) , <i>National School Organization Bangladesh</i>	2006-2011

Award & Honors

Fellowship: TTU Talkington End GRAD Sch , <i>Texas Tech University</i>	2026-2027
Fellowship: TTU Hutcheson Grad End AS Sch , <i>Texas Tech University</i>	2026-2027
Employee of The Month (2 Times) , <i>Bista Solutions Inc (Bangladesh Office)</i>	2024-2025
University Merit Scholarship , <i>Sylhet Engineering College, Shahjalal University of Science & Technology</i>	2016-2020
Bangladesh National Mathematics Olympiad , <i>Participant, Final Round, 2016</i>	2016
Inter University Robotics Competition , <i>North South University, 3rd Place</i>	2018
Honorable Mention for extraordinary academic and extracurricular activities , <i>Govt K.M.H. Degree College, Jhenaidah</i>	2015
National Talent Pool Scholarship, Junior School Certificate Exam , <i>Board of Intermediate & Secondary Education, Bangladesh</i>	2010
Prothom Alo Merit Scholarship Award , <i>Bangladesh, Board of Primary School Certificate, Prothom Alo, Bangladesh</i>	2007

Hackathons and Contests

- [1] Gemini API Developer Competition, Google, 2024 [\[Submission\]](#)
- [2] Hello GPT-4o AI Challenge, 2024 [\[Submission\]](#)[\[Certification\]](#)
- [3] Benin Multimodal AI Hackathon, 2024 [\[Submission\]](#)[\[Certification\]](#)
- [4] One Trillion Agents Hackathon 2025 [\[Submission\]](#)
- [5] COLAB Ventures: Bridging Business & Computing, 2025 [\[Submission\]](#)
- [6] DeveloperWeek 2025 Hackathon, 2025 [\[Submission\]](#)
- [7] CogX AI Fest-11Labs Hackathon, *Rank: 2nd Place (Silver Medal)*, 2024 [\[Certification\]](#)
- [8] TechnoVenza: Inter University Tech Contest, *Rank: 4th Place (App Development)*, 2019
- [9] SEC Inter University Programming Contest, *Rank: 9th place*, 2016
- [10] Bangladesh National Mathematics Olympiad: *Participant, Final Round*, 2016

OpenSource AI Models and Packages (Creator)

- | | | | |
|--|----------------|--|----------------|
| [1] AI Model: DeepSql-R1-Distil-8B [Link] | Download: 872. | [2] AI Model: SQLCoder-Qwen2.5-8bit [Link] | Download: 1515 |
| [3] AI Model: Text2OrmQuery-40B-8bit [Link] | Download: 254 | [4] AI Model: Odoo-ORM-v1-24B-fp32 [Link] | Download: 365 |
| [5] Package: MCPresso: Brew MCP Server Under 30 Seconds [Package] [Code] | | | |
| [6] Package: Agents-Driftguard: A State-Aware Rollback and Replanning Auto Agents [Package] [Code] | | | |

Technical Blogs

- [1] Adaptive Control: How Robots Can Adjust to Changing Environments [\[Link\]](#)
- [2] Kalman Filters: The Mathematics Behind Robot State Estimation [\[Link\]](#)
- [3] Reasoning-Intensive Retrieval in Agentic RAG: Dynamic, Reasoning-Aware Info Retrieval. [\[Link\]](#)
- [4] GraphRAG: Bridging the Gap Between Knowledge Graphs and GenAI for Improved Contextual Understanding [\[Link\]](#)
- [5] Protecting Privacy: Hide Sensitive Data Before Feeding It to LLMs. [\[Link\]](#)

Educational Courses Online Instructor

- | | |
|---|---|
| [1] Full-Stack Machine Learning and Data Science [Link] | [2] Communication System Lab Course [Link] |
| [3] Digital Electronic Lab Course [Link] | [4] Signals and Linear Systems Course [Link] |
| [5] Robotics Simulations Course [Link] | [6] Materials Science and Engineering Course [Link] |

References

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